Explainable Multi Task Deep Learning Framework for Automatic Spine Segmentation, Cobb Angle Estimation, and Scoliosis Severity Classification from X Ray Images

Authors

Name	Student ID
Mashrur Shakhawat Sayan	**1000060784**

Submission Date

25/06/2026

Advanced Training on Semiconductor and ICT Technology

BRAC University

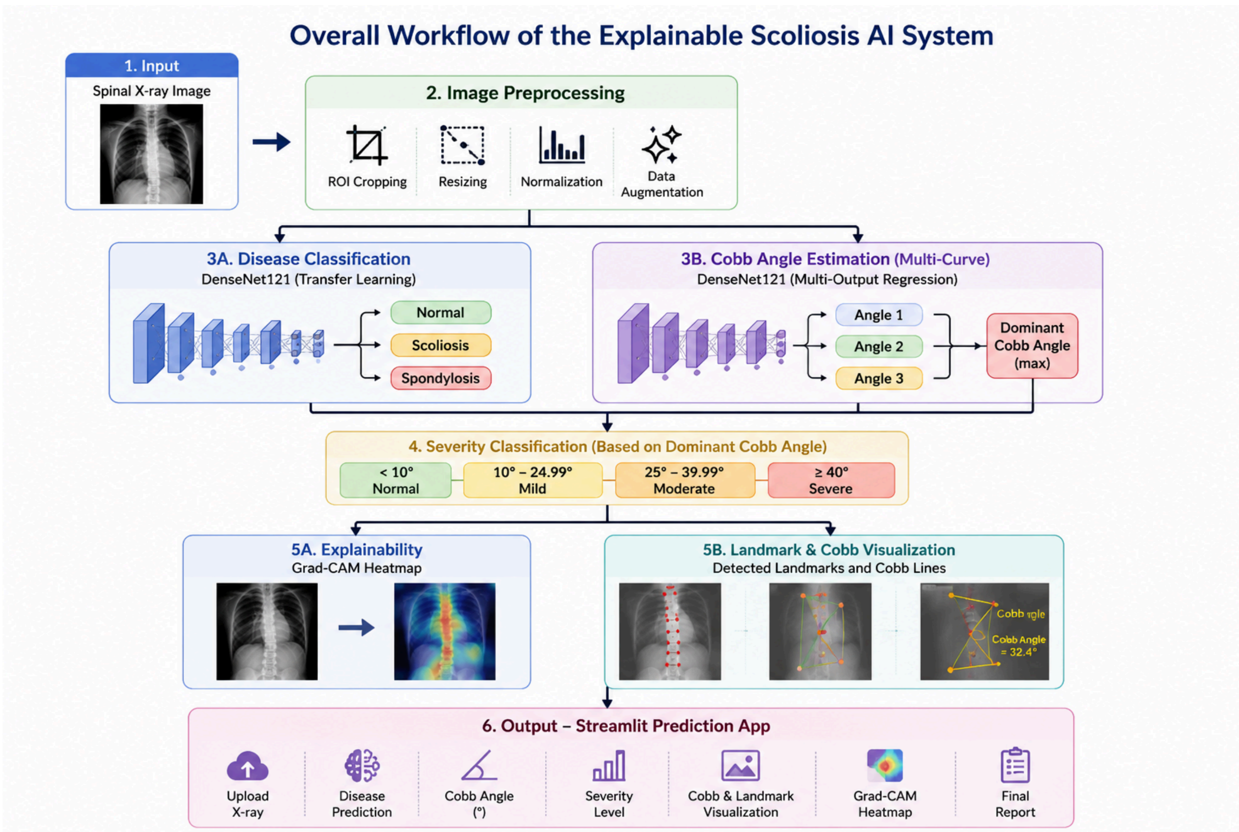
1. Introduction

1.1 Brief Introduction

Scoliosis is a spinal deformity characterized by an abnormal lateral curvature of the spine. Early diagnosis and accurate assessment of scoliosis severity are important for effective treatment planning and monitoring.

This project develops an explainable deep learning based medical imaging system that automatically analyzes spinal X ray images. The proposed system combines disease classification, Cobb angle estimation, scoliosis severity assessment, and explainable AI techniques within a unified framework.

The final system allows users to upload a spinal X ray image and receive an automated diagnosis, predicted Cobb angle, severity category, and visual explanations of the prediction.



1.2 Problem Statement

Manual scoliosis assessment requires experienced radiologists and orthopedic specialists to measure Cobb angles from X ray images. This process can be time consuming and subject to inter observer variability.

The proposed system receives a spinal X ray image as input and produces:

- Disease classification
- Cobb angle estimation
- Severity assessment
- Visual explanation of prediction

The system can be used by healthcare professionals, researchers, and medical students to assist scoliosis screening and evaluation.

Automated scoliosis assessment can improve efficiency, consistency, and accessibility of spinal health evaluation.

1.3 Objectives

The objectives of this project are:

1. Develop a deep learning model for spinal disease classification.
2. Estimate Cobb angles directly from spinal X ray images.
3. Determine scoliosis severity based on predicted Cobb angles.
4. Track experiments and model performance using MLflow.
5. Deploy the trained model through a Dockerized Streamlit application.
6. Ensure reproducibility through structured project organization and version control.

2. Dataset and Preparation

2.1 Dataset Description

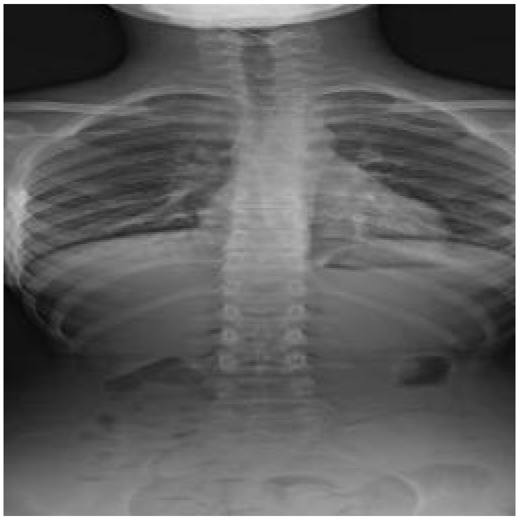
Two datasets were used in this project.

Dataset 1: Vertebrae X Ray Images Dataset

Source: Kaggle

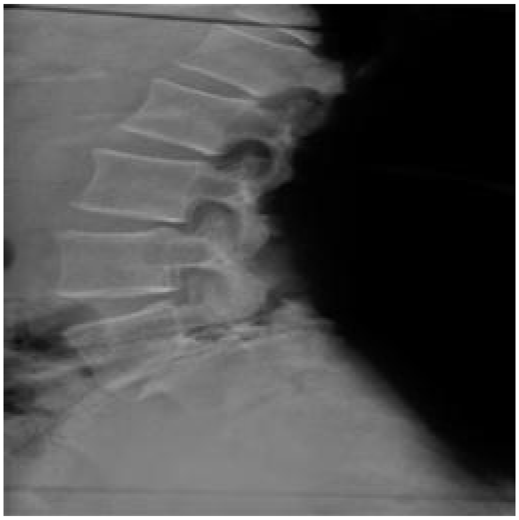
Classes:

- Normal



- Scoliosis

- Spondylosis(it is not the main focus in this project)



Total Images: 338

Input Type:

- Medical X ray Images

Target Variable:

- Disease Category

Dataset 2: AASCE MICCAI 2019 X Ray Dataset

Source: Kaggle

Samples:

- 481 spinal X ray images

Input Type:

- Medical X ray Images

Target Variables:

- Cobb Angle
- Vertebral Landmarks

2.2 Data Access

Datasets are downloaded automatically using Kaggle API scripts and stored locally inside the data directory.

The datasets are excluded from GitHub using .gitignore to comply with repository size and licensing requirements.

Only the download scripts are included in the repository.

2.3 Preprocessing

The following preprocessing steps were applied:

- Image resizing
- Image normalization
- Train validation test split
- Data augmentation
- Random horizontal transformations
- Rotation augmentation
- Landmark normalization
- Cobb angle label extraction

3. Methodology

3.1 Course Techniques Used

Technique	File/Module
Transfer Learning	src/train_classifier.py
CNN Based Classification	src/train_classifier.py
Multi Task Learning	src/train_cobb_multitask.py
Landmark Detection	src/train_landmark_detector.py
MLflow Experiment Tracking	src/train_classifier.py
Explainable AI (Grad CAM)	src/gradcam.py
Streamlit Deployment	app/app.py
Docker Deployment	Dockerfile

3.2 Model Architecture

Disease Classification Model

Backbone:

DenseNet121

Pretrained Weights:

ImageNet

Loss Function:

Cross Entropy Loss

Optimizer:

Adam

Learning Rate:

0.0001

Batch Size:

16

Multi Curve Cobb Regression Model

Backbone:

DenseNet121

Output Layer:

Three continuous angle outputs

Loss Function:

Mean Absolute Error Loss

Optimizer:

Adam

Landmark Detection Model

Backbone:

DenseNet121

Outputs:

- Vertebral Centers
- Endplate Landmarks
- Vertebral Ordering Information

3.3 Training Strategy

The models were trained using transfer learning with pretrained DenseNet121 weights.

Training included:

- Early stopping
- Learning rate scheduling
- Validation monitoring
- Model checkpointing

The best model was selected based on validation performance.

All experiments were logged using MLflow.

4. Experiments and Results

4.1 MLflow Tracking

MLflow was used to track all experiments.

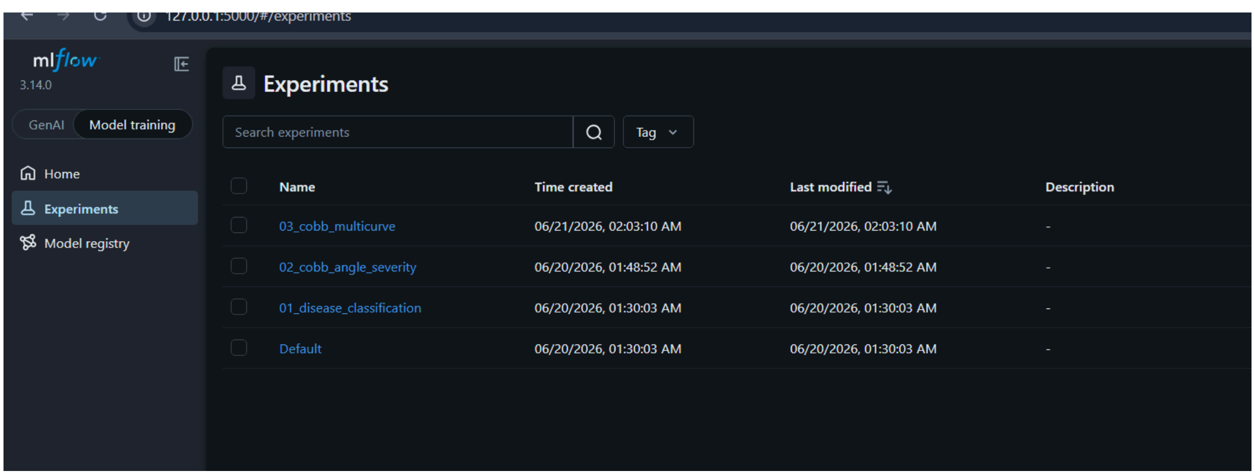
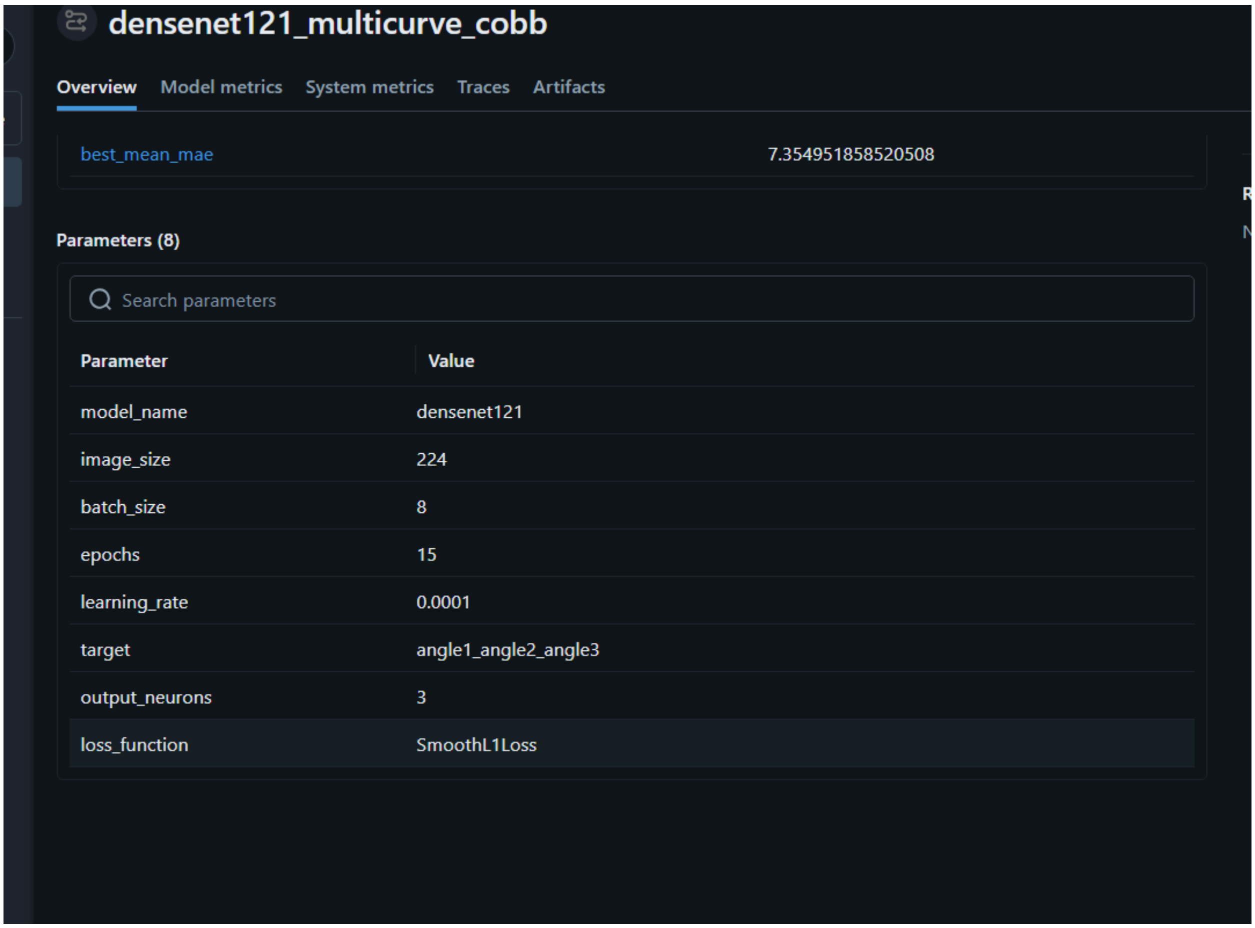


Figure 3. MLflow experiment dashboard showing all training experiments conducted for disease classification and Cobb angle estimation.

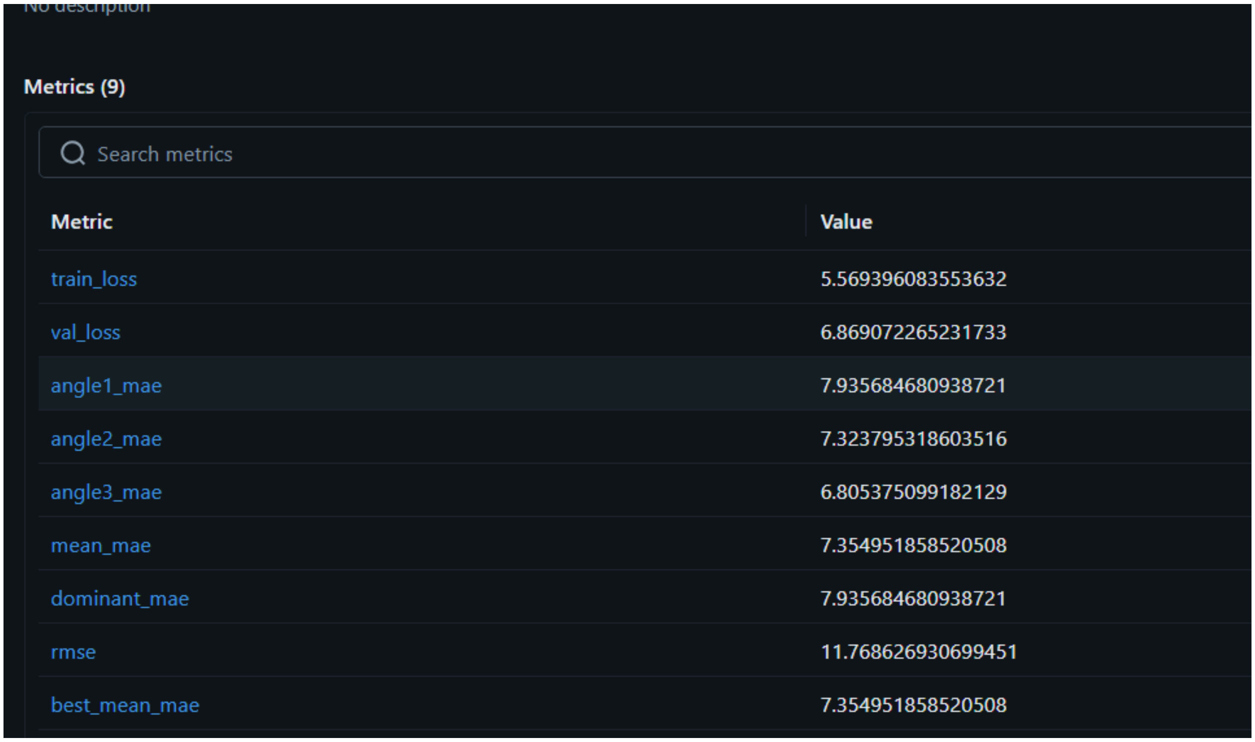
Parameters Logged:

- Learning Rate
- Batch Size
- Target
- Epochs



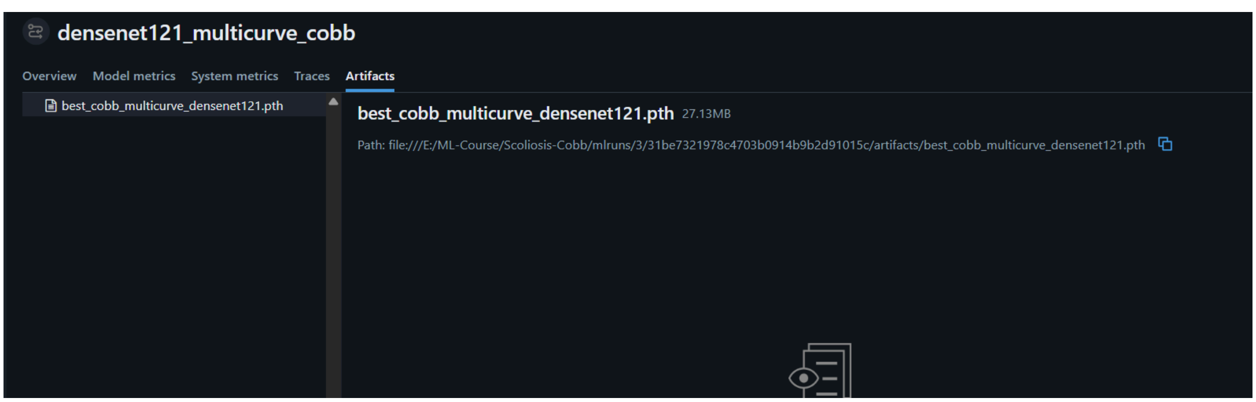
Metrics Logged:

- MAE
- Loss
- Angles Mae
- RMSE



Artifacts Saved

- Best Models
- Training Curves
- Evaluation Reports
- Confusion Matrices



Multiple training runs were recorded.

The best runs were selected according to validation accuracy and validation MAE.

4.2 Evaluation Results

Disease Classification

Metric	Value
Test Accuracy	98.53%
Precision	98.5%
Recall	98.5%
F1 Score	98.5%

Multi Curve Cobb Regression

Metric	Value
Mean Absolute Error	7.36°

Landmark Based Cobb Estimation

Metric	Value
MAE	3.41°
RMSE	5.03°
Median Error	2.97°

The evaluation results demonstrate excellent performance of the proposed system. The disease classification model achieved a test accuracy of **98.53%**, while the multicurve Cobb angle regression model obtained a mean absolute error of **7.36°**. The landmark based Cobb angle estimation also produced accurate measurements, achieving a mean absolute error of **3.41°**. These results indicate that the proposed framework can effectively classify spinal diseases and estimate Cobb angles with high accuracy.

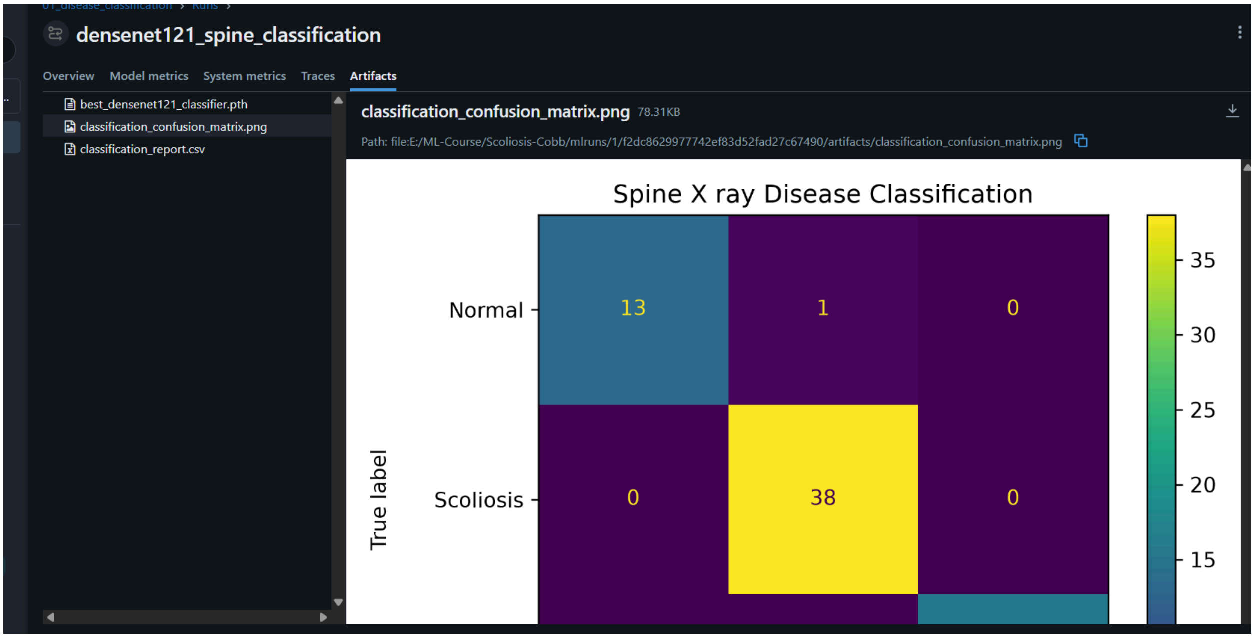
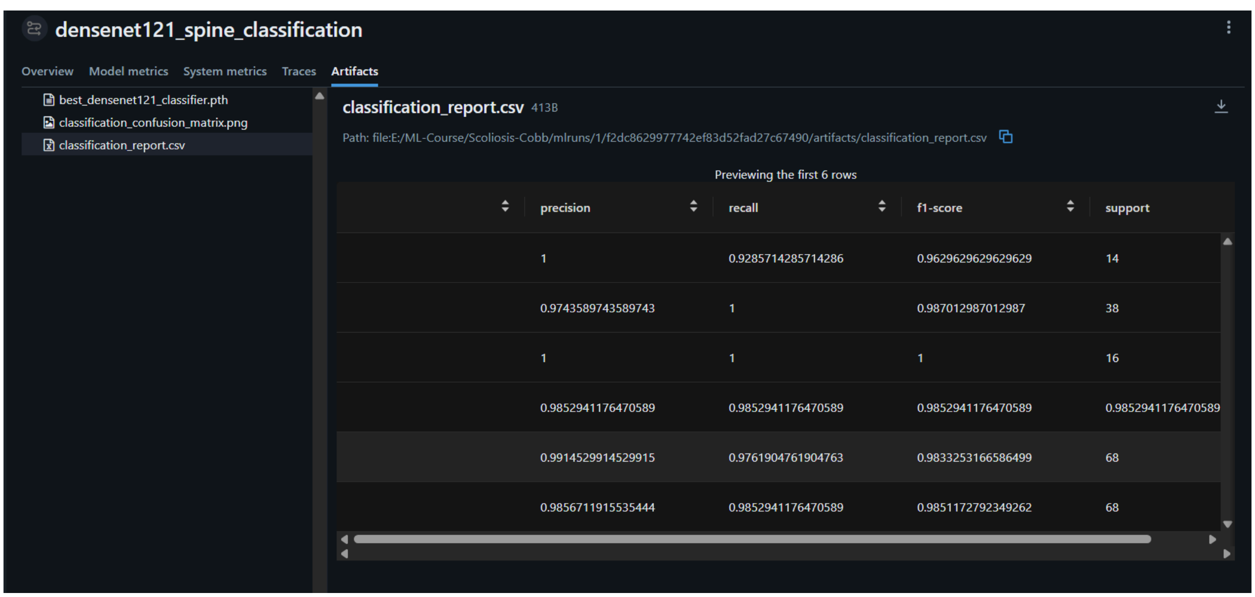


Figure 6. Confusion matrix of the DenseNet121 disease classification model evaluated on the test dataset.

The confusion matrix shows that the majority of X ray images were correctly classified into their respective disease categories, with only a few misclassifications. This demonstrates the robustness and reliability of the classification model.



	precision	recall	f1-score	support
	1	0.9285714285714286	0.9629629629629629	14
	0.9743589743589743	1	0.987012987012987	38
	1	1	1	16
	0.9852941176470589	0.9852941176470589	0.9852941176470589	0.9852941176470589
	0.9914529914529915	0.9761904761904763	0.9833253166586499	68
	0.9856711915535444	0.9852941176470589	0.9851172792349262	68

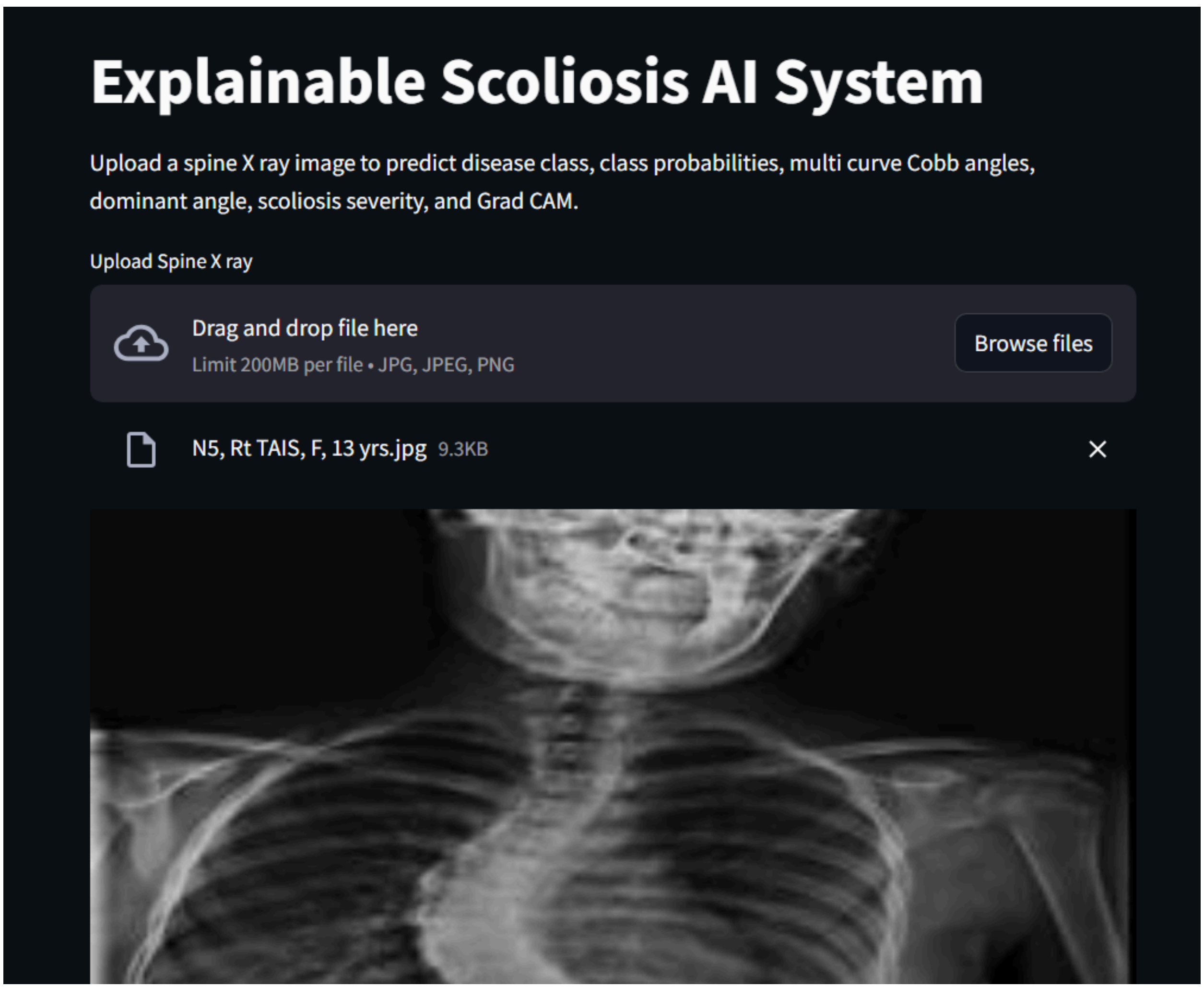
Figure 7. Classification report of the DenseNet121 disease classification model showing precision, recall, F1 score, and support for each disease category.

The classification report further confirms the model's strong performance, with high precision, recall, and F1 score across all classes, resulting in an overall classification accuracy of **98.53%**.

5. Prediction App and Docker

5.1 Prediction Pipeline

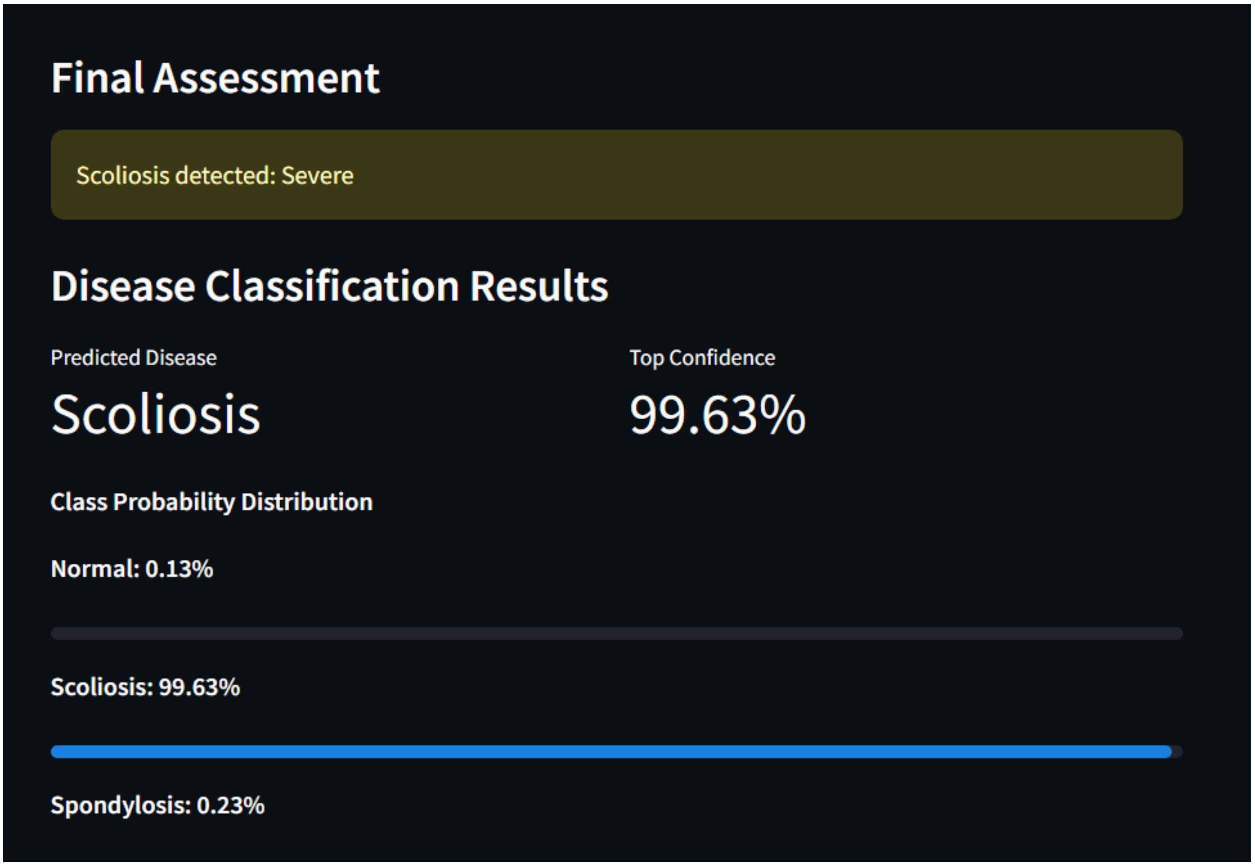
Users upload a spinal X ray image through the application.



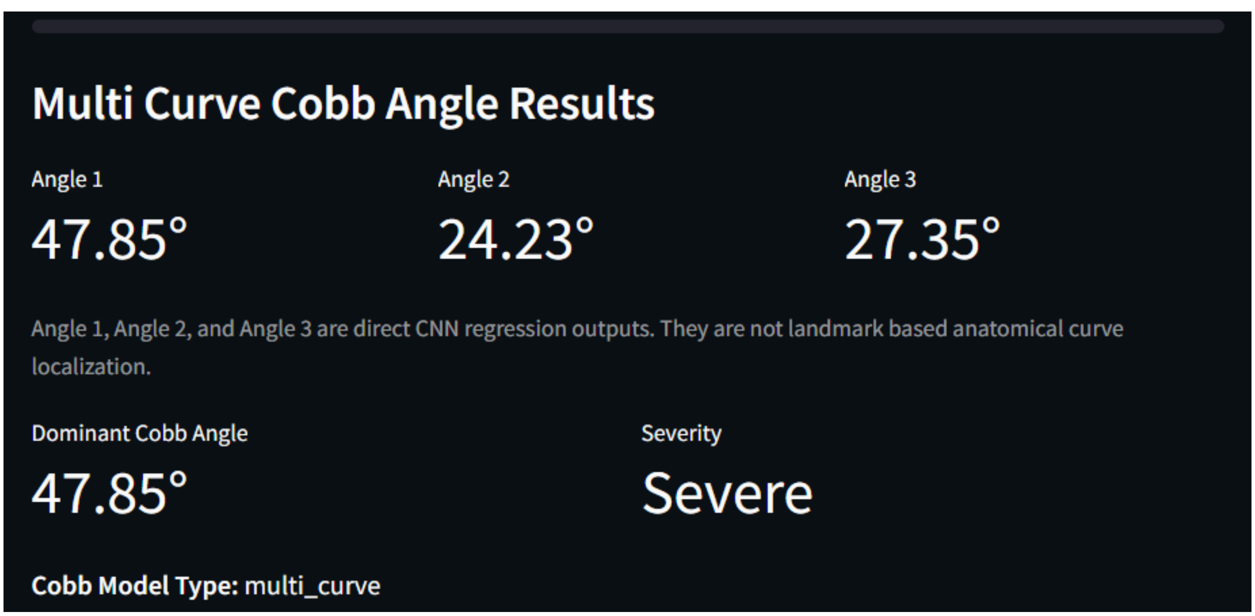
The image undergoes preprocessing and is passed through the trained models.

The system produces:

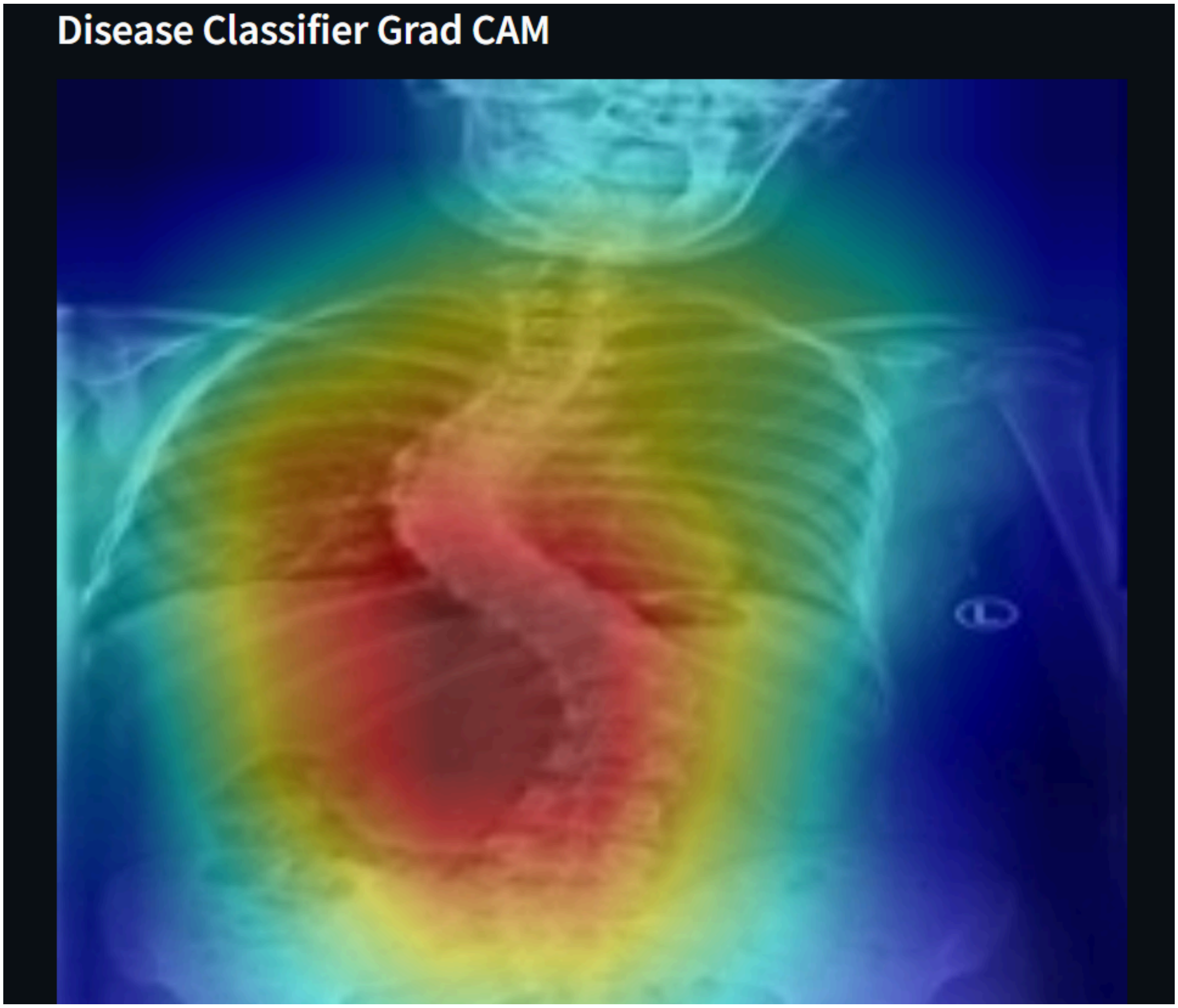
- Disease prediction



- Confidence score
- Cobb angle estimation



- Severity classification
- Explainability visualization



Severity levels are defined as:

- Normal: Less than 10°
- Mild: 10° to 24.99°
- Moderate: 25° to 39.99°
- Severe: Greater than or equal to 40°

5.2 Prediction UI

The application provides an intuitive interface for users to upload spinal X ray images in JPG, JPEG, or PNG format. After image upload, the application automatically preprocesses the input and displays the prediction results.

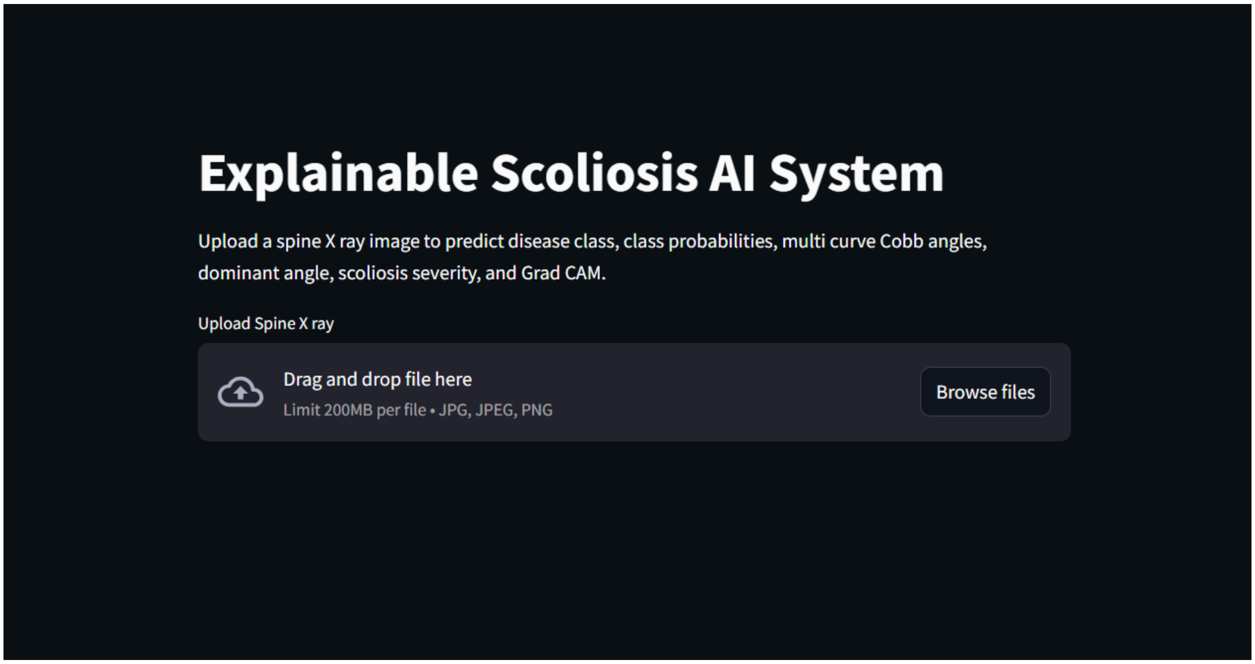


Figure 8. Homepage of the Explainable Scoliosis AI System developed using Streamlit, allowing users to upload spinal X ray images for automated analysis.

Features:

- Image Upload
- Disease Prediction
- Cobb Angle Prediction
- Severity Display
- Grad CAM Visualization

5.3 Docker Serving

The prediction application is containerized using Docker.

The repository contains:

- Dockerfile
- requirements.txt
- Deployment Instructions

Docker enables platform independent deployment and reproducibility.

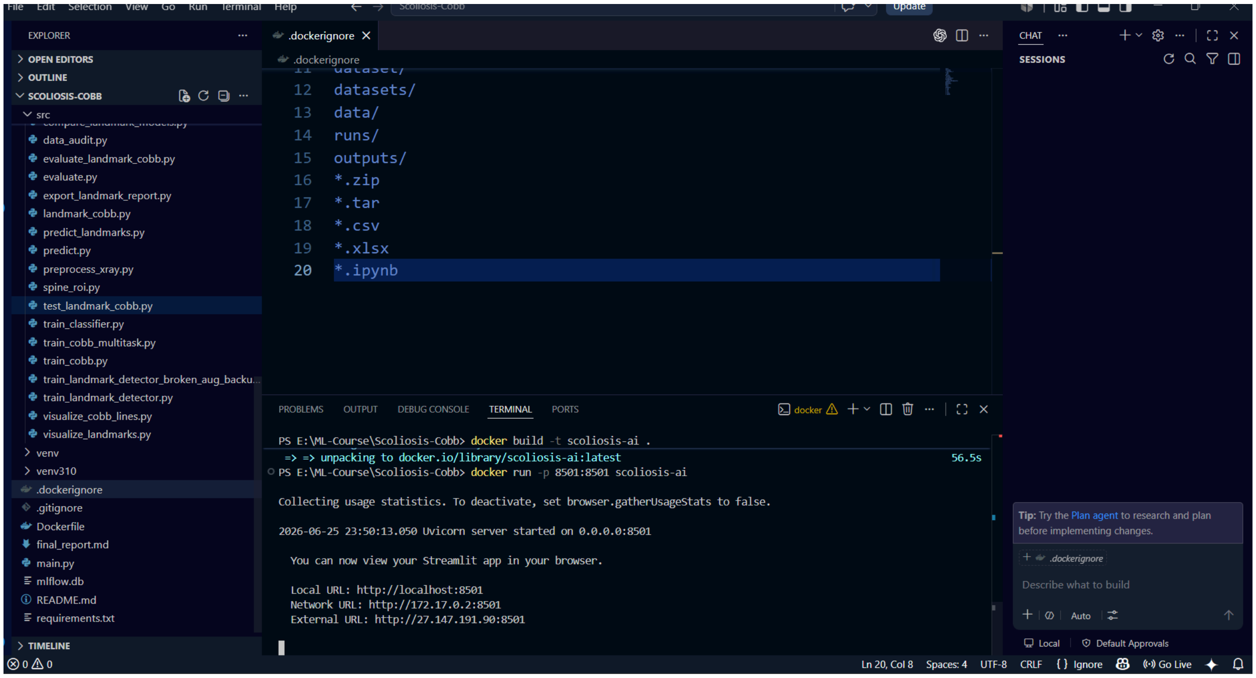


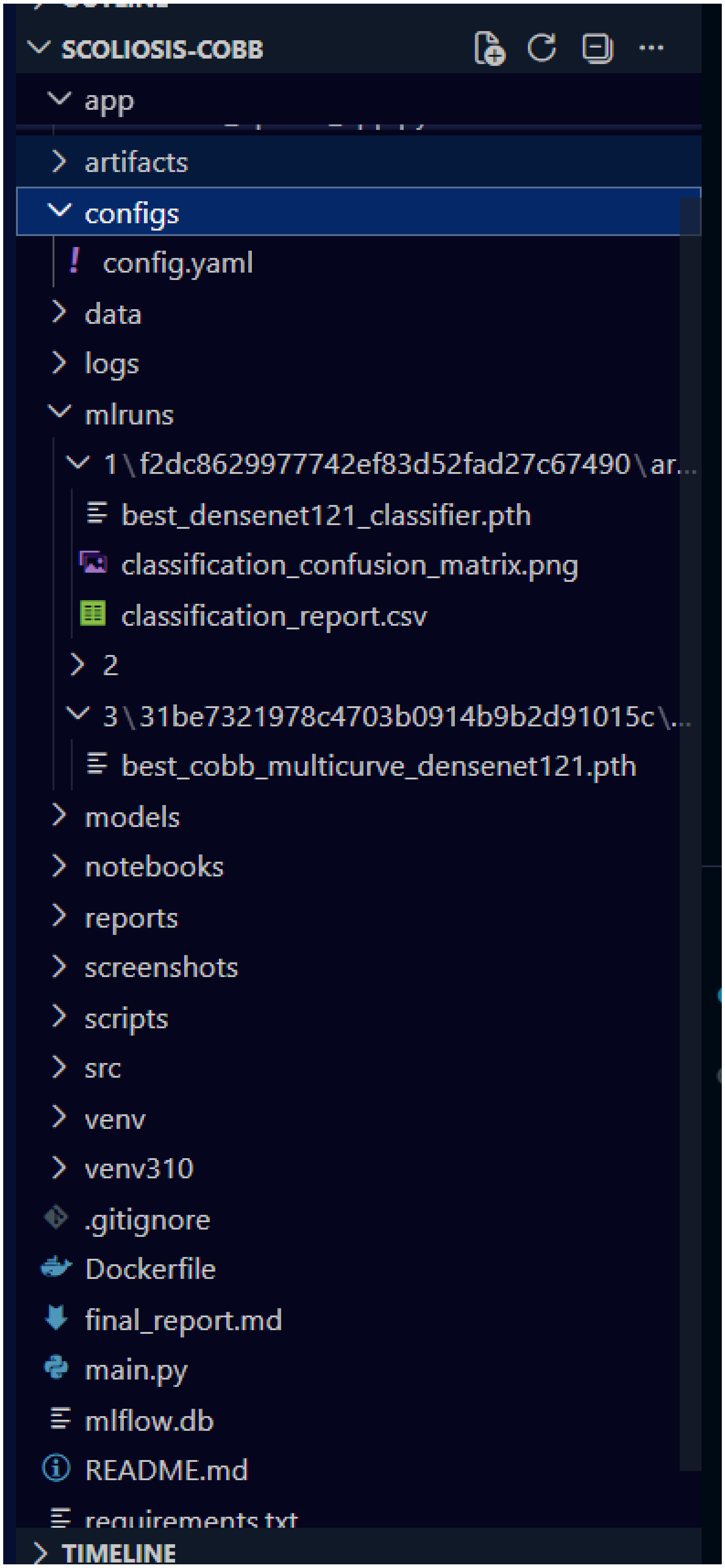
Figure 13. Dockerized deployment of the Explainable Scoliosis AI System showing successful execution of the Streamlit application inside a Docker container.

6. Repository and Reproducibility

6.1 Repository Structure

Scoliosis-Cobb/

├── app/
├── configs/
├── data/
├── models/
├── screenshots/
├── src/
├── mlruns/
├── Dockerfile
├── requirements.txt
├── README.md
├── final_report.md
└── .gitignore


SCOLIOSIS-COBB
app
artifacts
configs
config.yaml
data
logs
mlruns
1 \ f2dc8629977742ef83d52fad27c67490 \ ar...
best_densenet121_classifier.pth
classification_confusion_matrix.png
classification_report.csv
2
3 \ 31be7321978c4703b0914b9b2d91015c \ ...
best_cobb_multicurve_densenet121.pth
models
notebooks
reports
screenshots
scripts
src
venv
venv310
.gitignore
Dockerfile
final_report.md
main.py
mlflow.db
README.md
requirements.txt
TIMELINE

6.2 GitHub Rules

The following items are excluded:

- Raw datasets
- Virtual environments
- Cache files
- Credentials
- Large temporary files

The mlruns directory is included to preserve experiment history.

6.3 Reproducibility

A reviewer can reproduce the project by:

1. Installing dependencies.
2. Downloading datasets.
3. Training the models.
4. Reviewing MLflow runs.
5. Evaluating the models.
6. Running the Dockerized application.

7. Limitations and Future Work

7.1 Limitations

Current limitations include:

- Relatively small classification dataset
- Limited external validation
- Sensitivity to image quality variations
- Performance differences across unseen clinical data

7.2 Future Improvements

Future work may include:

- Larger multi center datasets

- Transformer based architectures
- Improved landmark localization
- Automatic segmentation integration
- Cloud deployment
- Enhanced explainability methods

8. Conclusion

This project presents an explainable deep learning framework for automatic scoliosis assessment from spinal X ray images.

The proposed system combines disease classification, Cobb angle estimation, severity assessment, explainable AI, MLflow experiment tracking, and Docker deployment within a single reproducible framework.

The classification model achieved 98.53% test accuracy, while the Cobb angle estimation models demonstrated promising performance for automated scoliosis evaluation.

The developed Streamlit application provides an accessible interface for end users and demonstrates the practical deployment of medical imaging AI systems.

The project provided valuable experience in deep learning, medical image analysis, experiment management, model deployment, and reproducible machine learning workflows.

9. References

[1] G. Huang, Z. Liu, L. Van Der Maaten, and K. Q. Weinberger, "Densely Connected Convolutional Networks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, USA, 2017, pp. 4700–4708.

[2] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad CAM: Visual Explanations from Deep Networks via Gradient based Localization," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, Venice, Italy, 2017, pp. 618–626.

[3] M. Zaharia et al., "Accelerating the Machine Learning Lifecycle with MLflow," *IEEE Data Engineering Bulletin*, vol. 41, no. 4, pp. 39–45, 2018.

[4] A. Vrtovec, D. Pernuš, B. Likar, and F. Ibragimov, "AASCE MICCAI 2019 Challenge: Automatic Assessment of Spinal Curvature from X Ray Images," *Medical Image Computing and Computer Assisted Intervention (MICCAI) Workshop*, 2019.